

The simplest geometric figures

Point, line and plane — the three words that are never defined, and everything built from them.

LESSON 1–2

2 × 40 MIN

MATEMATIKA 7, §1

S8 5

LESSON CLOCK

15'

25'

25'

20'

5'

Three undefined words	15 min
Notation	25 min
The axioms	25 min
Counting lines	20 min
Homework	5 min

LEARNING OBJECTIVES

- Name the point, the line and the plane as the undefined basic objects.
- Use the notation for points, lines, rays and segments correctly.
- State the axiom that two points determine exactly one line.
- Decide how many lines pass through a given set of points.

TEXTBOOK

National	pp. 3–8
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01

Explanation

Three undefined words

Geometry begins with three ideas that are **not** defined: the **point**, the **line** and the **plane**. Everything else is defined in terms of them.

OBJECT	WRITTEN	DRAWN AS
point	A, B, C	a dot
line	a, b or AB	a stroke with no ends
plane	α, β	a parallelogram

WHY NOT DEFINE THEM?

Every definition uses simpler words. Somewhere the chain has to stop, and mathematics chooses to stop at these three. Instead of definitions they are given **axioms** — statements accepted without proof that say how they behave.

Notation

FIGURE	WRITTEN	HAS
line AB	AB	no ends — it continues both ways
ray AB	$[AB)$	one end, at A
segment AB	$[AB]$	two ends
A lies on a	$A \in a$	—
A does not lie on a	$A \notin a$	—

A **ray** is half a line: it starts at a point and runs on forever in one direction. Two rays $[AB)$ and $[BA)$ on the same line point in opposite directions.

A LINE HAS NO ENDS AND NO LENGTH

Only a segment has a length. Asking for “the length of the line AB ” is asking for something that does not exist — the phrase must be “the length of the segment AB ”.

The axioms

AXIOM	STATEMENT
1	through any two points there passes exactly one line
2	two different lines meet in at most one point
3	on every line there are at least two points
4	a line divides the plane into two half-planes

Axiom 1 is the one used constantly. It is why a ruler needs only two marks to draw a line, and why two nails hold a plank rigid.

AXIOM 2 FOLLOWS FROM AXIOM 1

If two lines met in two points, then two points would have two lines through them — which Axiom 1 forbids. Deriving one statement from another is the whole activity of geometry, and it starts here.

Counting lines

How many lines pass through n points, no three of them on one line?

<i>N</i>	LINES	WHY
2	1	Axiom 1
3	3	each pair
4	6	$\frac{4 \times 3}{2}$
5	10	$\frac{5 \times 4}{2}$
<i>n</i>	$\frac{n(n-1)}{2}$	each pair, counted once

“NO THREE ON ONE LINE” IS ESSENTIAL

If all *n* points lie on one line, there is exactly **one** line, not $\frac{n(n-1)}{2}$. The condition is what makes the counting work.

02 Worked examples

EXAMPLE 1 How many lines pass through 5 points, no three on one line?

- 1

Each pair of points gives one line.
Axiom 1.
- 2

Pairs from 5: $\frac{5 \times 4}{2}$.
Each counted once.
- 3

$= 10$
- 4

All ten are different, since no three points are collinear.

ANSWER
10

EXAMPLE 2 Four points lie on a plane with exactly three of them on one line. How many lines are there?

- 1 The three collinear points give 1 line.
Not 3.
- 2 The fourth point joins each of the three: 3 more.
- 3 $1 + 3$
- 4 $= 4$ lines.

ANSWER

4

EXAMPLE 3 Name all the rays and segments determined by three points A, B, C on one line, in that order.

- 1 Segments: $[AB], [BC], [AC]$.
Three.
- 2 Rays from A : $[AB)$ — one direction only.
- 3 From B : $[BA)$ and $[BC)$.
- 4 From C : $[CB)$. Four rays in all.

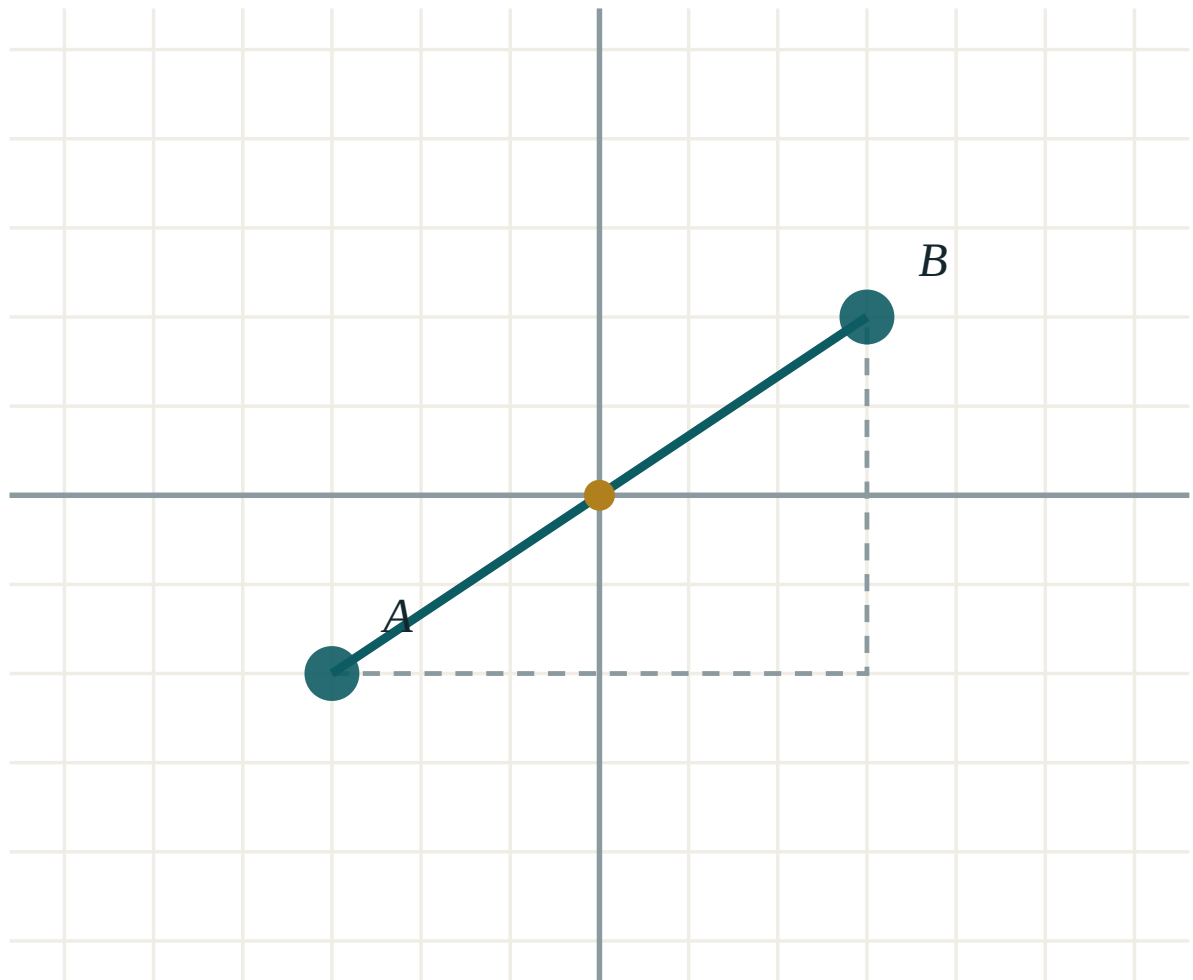
ANSWER

3 segments, 4 rays

03 Interactive model

Stretch a thread between two drawing pins and ask whether a second thread could take a different route; Axiom 1 is demonstrated before it is stated.

Points and the lines through them



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Point	Nuqta	Точка
Line	To'g'ri chiziq	Прямая
Plane	Tekislik	Плоскость
Ray	Nur	Луч
Segment	Kesma	Отрезок
Axiom	Aksioma	Аксиома
To belong to	Tegishli bo'lmoq	Принадлежать
Intersection	Kesishish	Пересечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which are undefined in geometry?

segment, ray, angle

point, line, plane

triangle, circle, square

none

Through two points there passes:

no line

exactly one line

two lines

many lines

A ray has:

no ends

one end

two ends

a length

Which has a length?

a line

a ray

a segment

a plane

5 points, no three collinear: how many lines?

5

10

20

25

Two different lines meet in:

no point

at most one point

two points

many points

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many lines through 2 points?

ANSWER 1

2. How many through 3 points, no three collinear?

ANSWER 3

3. How many through 4?

ANSWER 6

4. How many through 5?

ANSWER 10

5. Which figure has two ends?

ANSWER A segment

6. Which has one end?

ANSWER A ray

7. Which has none?

ANSWER A line

Medium · the standard question

7 PROBLEMS

1. How many through 6 points, no three collinear?

ANSWER 15

2. How many through 10?

ANSWER 45

3. Four points, three of them collinear

ANSWER 4 lines

4. Five points all on one line

ANSWER 1 line

5. Segments determined by 4 points on a line

ANSWER 6

6. Rays determined by 3 points on a line

ANSWER 4

7. Two lines meet in at most

ANSWER One point

Hard · stretch and reason

7 PROBLEMS

1. For which n do n points give 28 lines?

ANSWER $n = 8$

2. Five points with exactly four collinear

ANSWER 5 lines

3. Six points, three collinear and three collinear on another line

ANSWER 11 lines

4. How many segments have n points on a line as ends?

ANSWER $\frac{n(n-1)}{2}$

5. How many rays?

ANSWER $2(n-1)$

6. Why can two lines not meet twice?

ANSWER Axiom 1 would be broken

7. A plane is divided by 3 lines into at most

ANSWER 7 parts

07 Homework

Homework — 5 tasks

Draw a figure for every counting question; the picture is the proof.

- How many lines pass through 7 points, no three of them collinear?
- Four points lie in a plane, with exactly three on one line. How many lines do they determine?
- Name all the segments determined by four points on one line.
- State the four axioms of this lesson in your own words.

5. Explain why a line has no length but a segment does.